# Task 2: Multiple Sequence Alignment (MSA) of Hemoglobin Beta Proteins

## 1. Introduction

Multiple Sequence Alignment (MSA) is a bioinformatics technique used to align three or more biological sequences (protein or nucleic acid) to identify regions of similarity, conserved residues, evolutionary relationships, and functional motifs. In this study, Hemoglobin Beta (HBB) protein sequences from five different species were analyzed using Clustal Omega.

## 2. Objective

To perform Multiple Sequence Alignment of Hemoglobin Beta proteins from Human, Rat, Mouse, Sheep, and Chicken using Clustal Omega and identify conserved regions and evolutionary relationships among the selected species.

## 3. Protein Sequences Used

• Human (Homo sapiens) – Hemoglobin Beta (HBB)
• Rat (Rattus norvegicus) – Hemoglobin Beta (HBB)
• Mouse (Mus musculus) – Hemoglobin Beta (HBB)
• Sheep (Ovis aries) – Hemoglobin Beta (HBB)
• Chicken (Gallus gallus) – Hemoglobin Beta (HBB)

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>sp|P68871|HBB_HUMAN Hemoglobin subunit beta OS=Homo sapiens OX=9606 GN=HBB PE=1 SV=2
MVHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAVMGNPK
VKAHGKKVLGAFSDGLAHLDNLKGTFATLSELHCDKLHVDPENFRLLGNVLVCVLAHHFG
KEFTPPVQAAYQKVVAGVANALAHKYH


>sp|P11517|HBB2_RAT Hemoglobin subunit beta-2 OS=Rattus norvegicus OX=10116 PE=1 SV=2
MVHLTDAEKATVSGLWGKVNADNVGAEALGRLLVVYPWTQRYFSKFGDLSSASAIMGNPQ
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KEFTPCAQAAFQKVVAGVASALAHKYH


>sp|P02089|HBB2_MOUSE Hemoglobin subunit beta-2 OS=Mus musculus OX=10090 GN=Hbb-b2 PE=1 SV=2
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>sp|P02075|HBB_SHEEP Hemoglobin subunit beta OS=Ovis aries OX=9940 GN=HBB PE=1 SV=2
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>sp|P02112|HBB_CHICK Hemoglobin subunit beta OS=Gallus gallus OX=9031 GN=HBB PE=1 SV=2
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VRAHGKKVLTSFGDAVKNLDNIKNTFSQLSELHCDKLHVDPENFRLLGDILIIVLAAHFS
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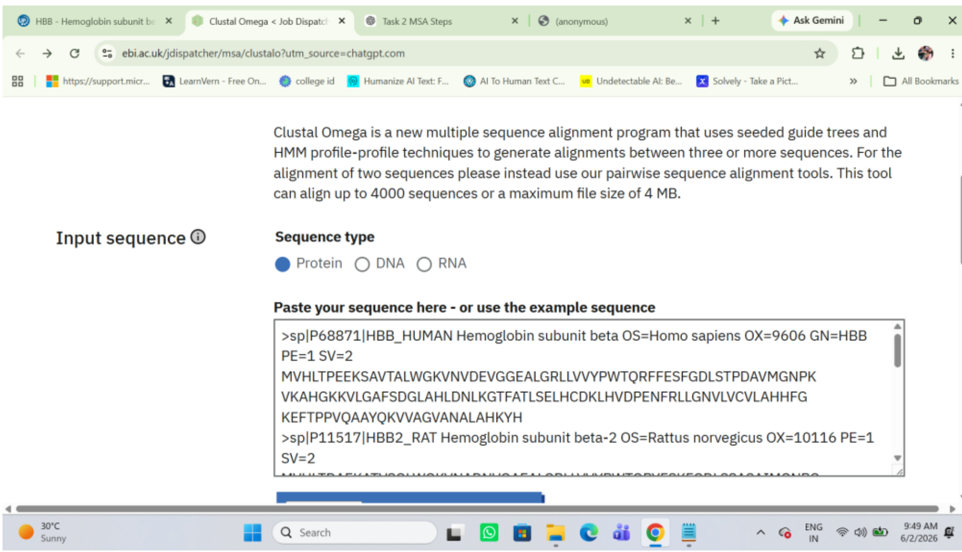
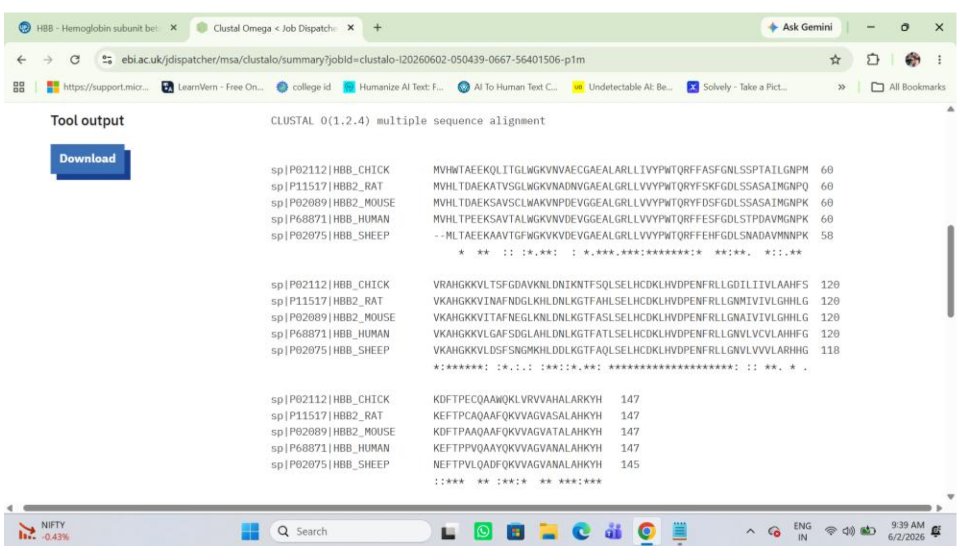
## 4. Methodology

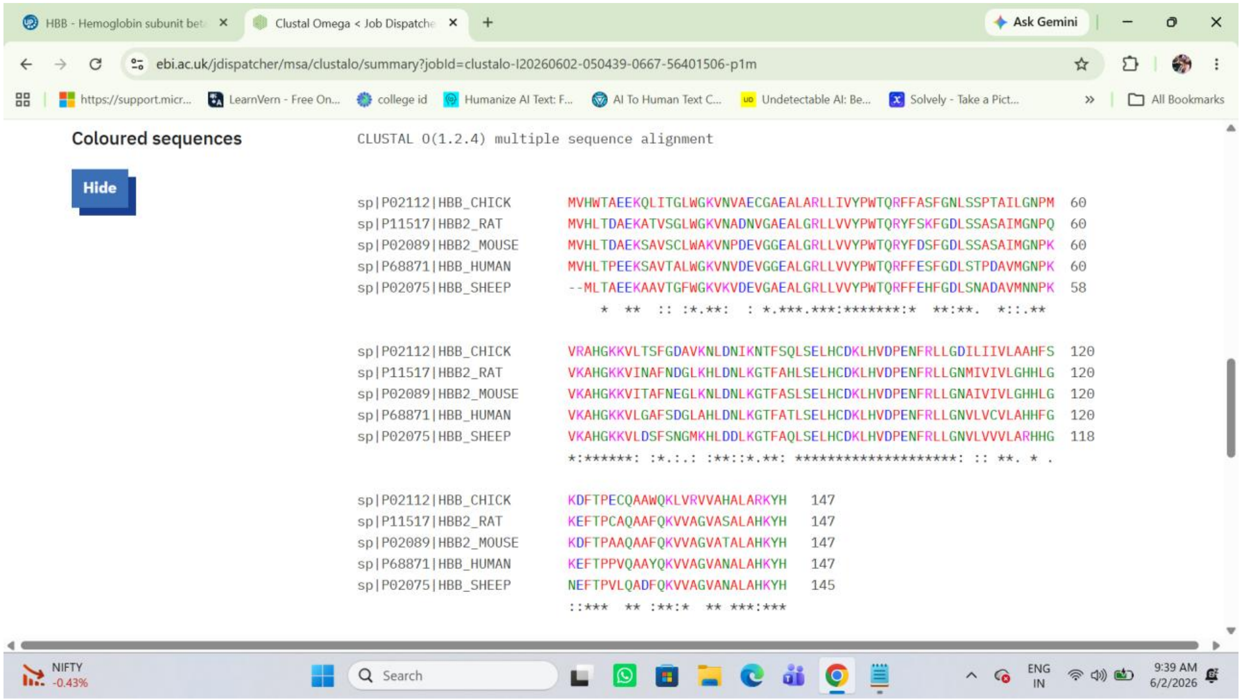
1. Downloaded Hemoglobin Beta protein sequences from UniProt.
2. Combined all sequences into a single FASTA file.
3. Performed Multiple Sequence Alignment using Clustal Omega.
4. Analyzed conserved residues and motifs.
5. Generated Guide Tree and Phylogenetic Tree.
6. Interpreted evolutionary relationships.




## 5. Results and Analysis

The alignment revealed several conserved amino acid residues across all species. In Clustal Omega output, '*' indicates fully conserved residues, ':' indicates strong similarity, and '.' indicates weak similarity. The presence of many conserved positions indicates the functional importance of Hemoglobin Beta.

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CLUSTAL O(1.2.4) multiple sequence alignment
sp|P02112|HBB_CHICK      MVHWTAEEKQLITGLWGKVNVAECGAEALARLLIVYPWTQRFFASFGNLSSPTAILGNPM 60
sp|P11517|HBB2_RAT       MVHLTDAEKATVSGLWGKVNADNVGAEALGRLLVVYPWTQRYFSKFGDLSSASAIMGNPQ 60
sp|P02089|HBB2_MOUSE     MVHLTDAEKSAVSCLWAKVNPDEVGGEALGRLLVVYPWTQRYFDSFGDLSSASAIMGNPK 60
sp|P68871|HBB_HUMAN      MVHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAVMGNPK 60
sp|P02075|HBB_SHEEP      --MLTAEEKAAVTGFWGKVKVDEVGAEALGRLLVVYPWTQRFFEHFGDLSNADAVMNNPK 58
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sp|P02112|HBB_CHICK      VRAHGKKVLTSFGDAVKNLDNIKNTFSQLSELHCDKLHVDPENFRLLGDILIIVLAAHFS 120
sp|P11517|HBB2_RAT       VKAHGKKVINAFNDGLKHLDNLKGTFAHLSELHCDKLHVDPENFRLLGNMIVIVLGHHLG 120
sp|P02089|HBB2_MOUSE     VKAHGKKVITAFNEGLKNLDNLKGTFASLSELHCDKLHVDPENFRLLGNAIVIVLGHHLG 120
sp|P68871|HBB_HUMAN      VKAHGKKVLGAFSDGLAHLDNLKGTFATLSELHCDKLHVDPENFRLLGNVLVCVLAHHFG 120
sp|P02075|HBB_SHEEP      VKAHGKKVLDSFSNGMKHLDDLKGTFAQLSELHCDKLHVDPENFRLLGNVLVVVLARHHG 118
                         *:******: :*.:.: :**::*.**: *******************: :: **. * .
sp|P02112|HBB_CHICK      KDFTPECQAAWQKLVRVVAHALARKYH   147
sp|P11517|HBB2_RAT       KEFTPCAQAAFQKVVAGVASALAHKYH   147
sp|P02089|HBB2_MOUSE     KDFTPAAQAAFQKVVAGVATALAHKYH   147
sp|P68871|HBB_HUMAN      KEFTPPVQAAYQKVVAGVANALAHKYH   147
sp|P02075|HBB_SHEEP      NEFTPVLQADFQKVVAGVANALAHKYH   145
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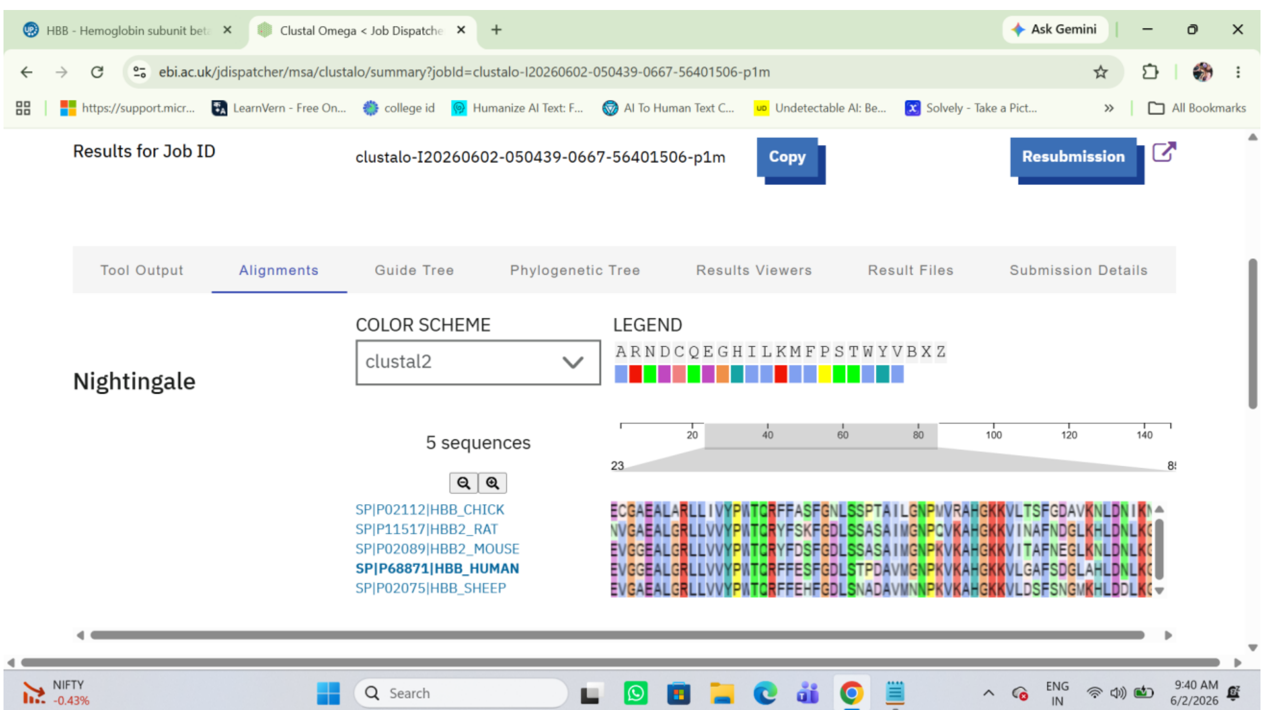
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SP|P02112|HBB_CHICK
SP|P11517|HBB2_RAT
SP|P02089|HBB2_MOUSE
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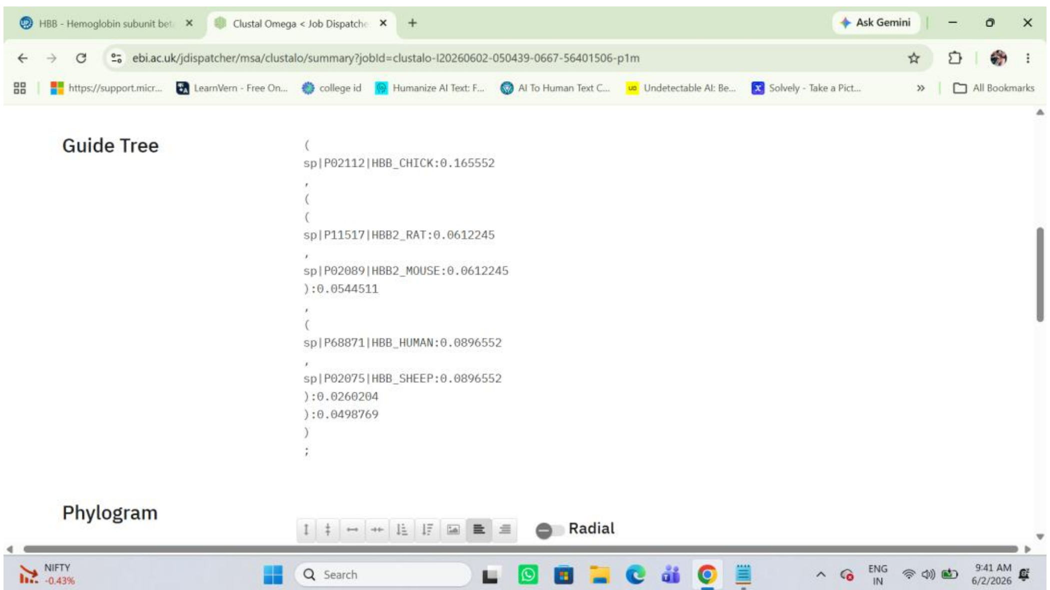
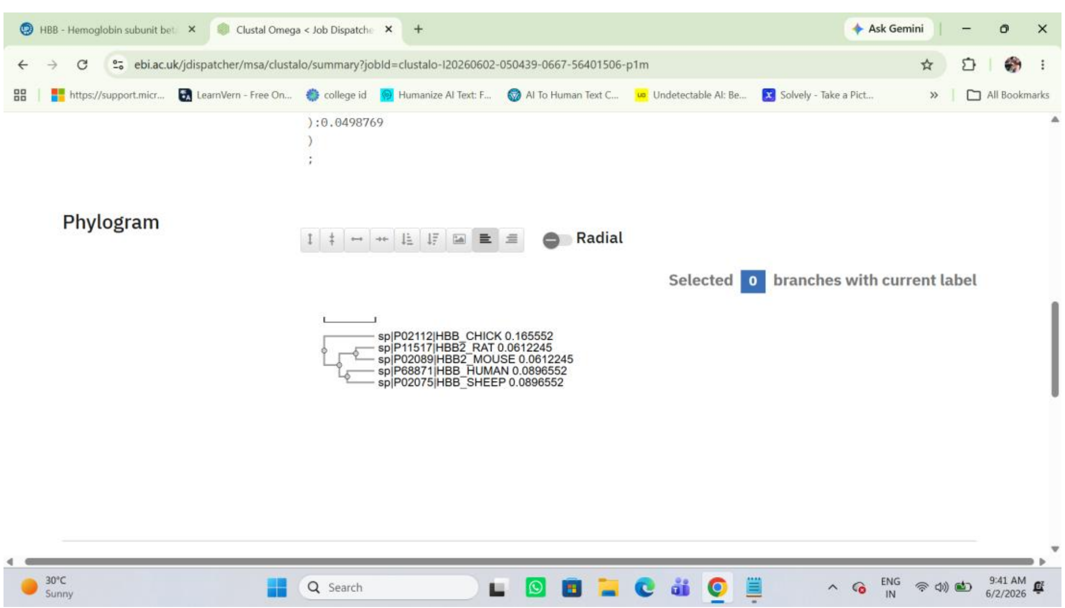
## 6. Conserved Region Interpretation

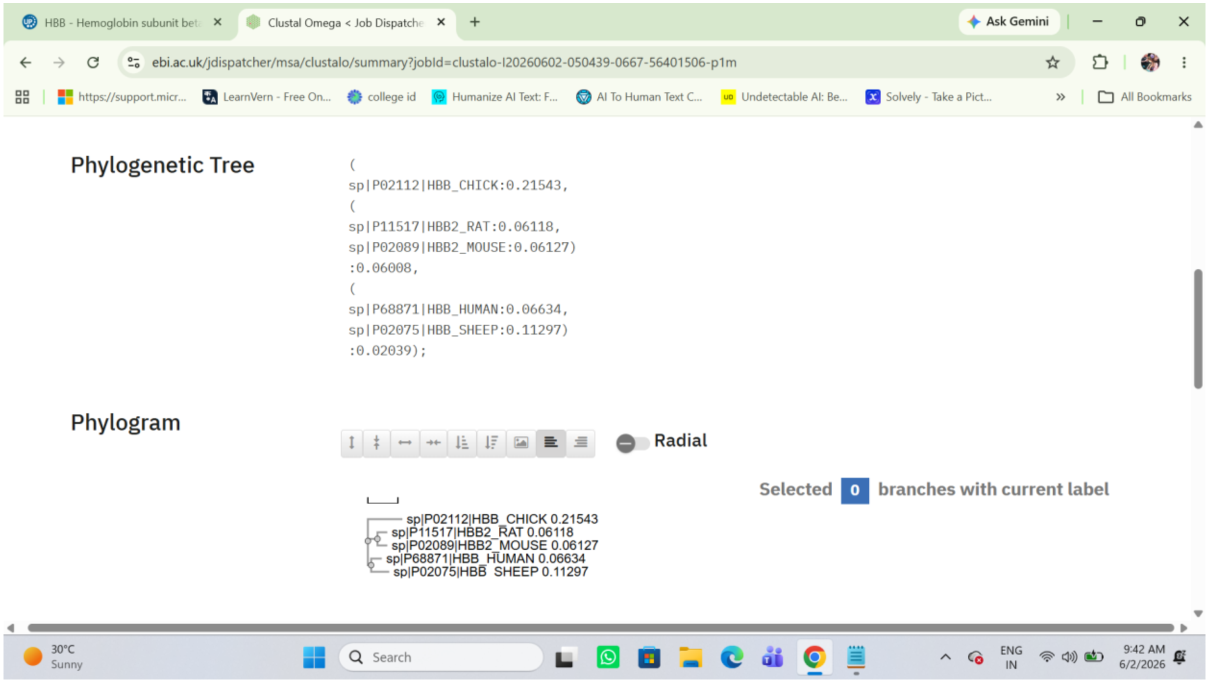
The presence of numerous conserved amino acids demonstrates that critical structural and functional regions of Hemoglobin Beta have been preserved throughout evolution. These residues contribute to oxygen transport and protein stability.

## 7. Guide Tree and Phylogenetic Analysis

Mouse and Rat formed the closest cluster, indicating a close evolutionary relationship. Human and Sheep also showed high sequence similarity. Chicken formed a separate branch and appeared more distant from the mammalian species. This suggests that mammalian hemoglobin proteins are more closely related to each other than to avian hemoglobin proteins.






## 8. Discussion

The results demonstrate that Hemoglobin Beta proteins remain highly conserved because of their essential biological function. Variations observed among species likely represent evolutionary adaptations while preserving core functionality.

## 9. Conclusion

Multiple Sequence Alignment of Hemoglobin Beta proteins from Human, Rat, Mouse, Sheep, and Chicken was successfully performed using Clustal Omega. Conserved amino acid residues and motifs were identified, highlighting the functional importance of Hemoglobin Beta. Phylogenetic analysis revealed that Mouse and Rat are the most closely related species, whereas Chicken is evolutionarily more distant.